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SOLUTION COLLECTION DEVICE WITH EVALUATION ELEMENT

Abstract

A blood testing device for detecting hemolysis in a blood sample is described. The blood testing device comprises an housing for containing the blood sample. The housing has a treatment window and an optical zone formed therein. The blood testing device further includes an acoustic transducer positioned to selectively generate acoustic forces directed into the treatment window of the housing and a control unit for selectively actuating and deactuating the acoustic transducer to permit colorimetric analysis of plasma within the blood sample.

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Background/Summary

[0001] This application is a continuation of U.S. Ser. No. 17/299,816, filed Jun. 4, 2021; which is a U.S. National Stage application filed under 35 USC § 371 of International Application No. PCT/US2019/064623, filed Dec. 5, 2019; which claims benefit under 35 USC § 119 (e) of U.S. Provisional Application No. 62/776,825, filed Dec. 7, 2018. The entire contents of each of the above-referenced patent applications are hereby expressly incorporated herein by reference.

BACKGROUND OF THE INVENTION

[0002] Point-of-care testing refers generally to medical testing at or near the site of patient care, such as in an emergency room. A desired outcome of such tests is often rapid and accurate lab results to determine a next course of action in the patient care. A number of such point of care tests involve analysis of a blood sample from the patient. The ideal blood sample is pure plasma separated from the source whole blood sample. However, even in such plasma samples, there are often residual broken blood cells as a result of hemolysis due to imperfections in obtaining the sample from the subject, pre-analytical blood sample handling, and the whole blood separation process. In certain cases, these hemolysed cells can interfere with the integrity of analytical test results.

[0003] For example, if hemolysis occurs, resulting free hemoglobin in the sample may cause interference in a number of tests, thereby leading to a signal reduction, reduced measurement accuracy and precision, or to false positive results at the other end of the spectrum. For one, it has been found that the potassium concentration in a corresponding sample may increase significantly and cause a high risk of misdiagnosis in a diagnostic test for potassium levels.

[0004] To determine whether hemolysis has occurred, a number of tests have been developed to determine hemoglobin (Hb) levels in a blood sample. One common reagent used for determining Hb levels or hemolysis in a blood sample is referred to as Drabkin's Reagent. Drabkin's Reagent comprises a mixture that works by lysing red blood cells and quantitatively converting all Hb in a sample into one form, cyanomethaemoglobin, which is then be measured on a spectrometer using a single wavelength. As such, Drabkin's Reagent measures intracellular hemoglobin as well as free hemoglobin. For at least this reason, Drabkin's Reagent does not provide a realistic picture of the extent of free Hb present at a particular point in time in a sample, which is indicative of hemolysis.

[0005] A need exists, therefore, for rapid, point-of-care testing of a blood sample to determine whether hemolysis has occurred that overcomes the shortcomings of the present testing regimes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate one or more implementations described herein and, together with the description, explain these implementations. In the drawings:

[0007] FIG. 1 is a cross sectional view of a portion of a blood testing device constructed in

accordance with one embodiment of the present disclosure containing a blood sample.

[0008] FIG. 2 is a cross sectional view of a portion of the blood testing device of FIG. 1 showing a sample of acoustically treated blood in accordance with one embodiment of the present disclosure in which blood cells and plasma within a blood sample are separated into a first zone containing plasma and blood cells, and a second zone containing plasma and being substantially devoid of blood cells.

[0009] FIG. 3 is an orthogonal view of a fluid collection device having an integrated blood testing device constructed in accordance with the present disclosure.

[0010] FIGS. 4A-4C illustrate views of portions of another version of a fluid collection device constructed in accordance with the present disclosure.

[0011] FIG. 5A illustrates a fluid housing of FIG. 4 being inserted into a slot of a reader for acoustic treatment and colorimetric analysis in accordance with one embodiment of the present disclosure.

[0012] FIG. 5B illustrates the fluid housing of FIG. 4 being inserted into a port of another version of a reader for acoustic treatment and colorimetric analysis in accordance with one embodiment of the present disclosure.

[0013] FIG. 6 illustrates an exploded, perspective view of a blood testing device removably attachable to a fluid housing such as a syringe constructed in accordance with one embodiment of the present disclosure.

[0014] FIG. 7 illustrates a perspective view of another version of a blood testing device constructed in accordance with one embodiment of the present disclosure.

[0015] FIG. 8 illustrates top plan view of a blood testing device constructed in accordance with one embodiment of the present disclosure.

[0016] FIG. 9 illustrates an exploded, perspective view of a blood testing device removably attachable to a fluid housing such as a syringe constructed in accordance with one embodiment of the present disclosure.

[0017] FIG. 10 illustrates a perspective view of another version of a blood testing device constructed in accordance with one embodiment of the present disclosure.

[0018] FIG. 11 illustrates a perspective view of a blood testing device having a closeable gate constructed in accordance with one embodiment of the present disclosure.

[0019] FIG. 12 illustrates a cross-sectional view of the blood testing device of FIG. 11 having the closeable gate in a first position, and positioned within a slot of a reader in accordance with the present disclosure.

[0020] FIG. 13 illustrates another cross-sectional view of the blood testing device of FIG. 11 having the closeable gate in a second position, and positioned within the slot of the reader in accordance with the present disclosure.

[0021] FIG. 14 illustrates a perspective view of another version of a blood testing device having a moveable gate constructed in accordance with one embodiment of the present disclosure.

[0022] FIG. 15 illustrates a cross-sectional view of the blood testing device of FIG. 14 having the moveable gate in a first position, and positioned within a slot of the reader in accordance with the present disclosure.

[0023] FIG. 16 illustrates another cross-sectional view of the blood testing device of FIG. 14 having the moveable gate in a second position, and positioned within the slot of the reader in accordance with the present disclosure.

DETAILED DESCRIPTION

[0024] The following detailed description refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements.

[0025] As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily

limited to only those elements, but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

[0026] In addition, use of the “a” or “an” are employed to describe elements and components of the embodiments herein. This is done merely for convenience and to give a general sense of the inventive concept. This description should be read to include one or more and the singular also includes the plural unless it is obvious that it is meant otherwise.

[0027] Further, use of the term “plurality” is meant to convey “more than one” unless expressly stated to the contrary.

[0028] As used herein any reference to “one embodiment” or “an embodiment” means that a particular element, feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment.

[0029] Circuitry, as used herein, may be analog and/or digital, components, or one or more suitably programmed microprocessors and associated hardware and software, or hardwired logic. Also, “components” may perform one or more functions. The term “component,” may include hardware, such as a processor, an application specific integrated circuit (ASIC), or a field programmable gate array (FPGA), or a combination of hardware and software. Software includes one or more computer executable instructions that when executed by one or more component cause the component to perform a specified function. It should be understood that the algorithms described herein are stored on one or more non-transitory memory. Exemplary non-transitory memory includes random access memory, read only memory, flash memory or the like. Such non-transitory memory may be electrically based or optically based.

[0030] As used herein, the term “substantially” means that the subsequently described parameter, event, or circumstance completely occurs or that the subsequently described parameter, event, or circumstance occurs to a great extent or degree. For example, the term “substantially” means that the subsequently described parameter, event, or circumstance occurs at least 90% of the time, or at least 91%, or at least 92%, or at least 93%, or at least 94%, or at least 95%, or at least 96%, or at least 97%, or at least 98%, or at least 99%, of the time, or means that the dimension or measurement is within at least 90%, or at least 91%, or at least 92%, or at least 93%, or at least 94%, or at least 95%, or at least 96%, or at least 97%, or at least 98%, or at least 99%, of the referenced dimension or measurement.

[0031] In accordance with one aspect, there are provided devices, systems, and processes for determining a presence of hemolysis in a sample. Advantageously, devices, systems, and processes described herein determine whether hemolysis has occurred in a sample based upon a colorimetry assessment of a portion of the sample.

[0032] In accordance with another aspect, there are provided devices, systems, and processes for a blood collection container having a hemolysis indicating feature.

[0033] In accordance with another aspect, there are provided blood collection devices, systems, accessories and processes having a plasma separating feature.

[0034] In accordance with another aspect, there are provided blood collection devices, systems, accessories, and processes having a hemolysis indicating feature.

[0035] Referring now to the Figures and in particular to FIG. 1, shown therein is a diagrammatic view of a blood testing device **10** constructed in accordance with the present disclosure. In general, the blood testing device **10** includes a housing **12**, an acoustic transducer **16**, a reader **18**, and a control unit **20** connected to the transducer **16** and the reader **18**. The housing **12** is constructed of a fluid impermeable material so that the housing **12** can hold and contain a sample of blood containing blood cells, suspended within plasma. The housing **12** can be a syringe or a vacutainer,

for example that can be used for collecting blood and transporting the blood for purposes of testing. The blood may be collected from an animal, such as a human, or a non-human (such as a cat, dog, cow, horse, fish, or the like). The acoustic transducer **16**, the reader **18**, and the control unit **20** may be located outside of the housing **12** as shown in FIG. **1** or inside the housing **12** (not shown). The acoustic transducer **16** selectively generates acoustic forces that are directed to the housing **12**. In some embodiments, the acoustic transducer **16** can be tuned so as to provide a magnitude and/or frequency of acoustic forces so as to facilitate separation of the undamaged blood cells from the plasma and damaged blood cells. The magnitude and/or frequency of the acoustic forces generated by the acoustic transducer **16** can be selected depending upon a size and/or construction of the housing **12**, or composition of the blood sample within the housing **12**. In one embodiment, the acoustic transducer **16** can be a piezoelectric element. At least a portion of the housing **12**, adjacent to the transducer **16**, is constructed of a material that functions to pass the acoustic forces generated by the acoustic transducer **16** into the sample contained within the housing **12**. Exemplary materials that can be used to form the housing **12** include glass, crystal, and the like. Parts of the housing **12** away from the transducer **16** can be made of other materials such as plastic. The application of the acoustic forces into the sample by the acoustic transducer **16** causes the blood cells within the blood to move within the plasma to form a first zone **22** having an increased density or concentration of the blood cells than the blood contained prior to the application of the acoustic forces, and at least one second zone **24** being substantially only plasma, i.e., substantially devoid of any undamaged blood cells. The reader **18** is positioned adjacent to the second zone **24** and functions to read at least one parameter of the plasma. In one embodiment, the reader **18** is an optical reader, such as a camera or photospectrometer having a field of view overlapping with the housing **12** such that the plasma within the second zone is visible to the reader **18**. The optical reader **18** is positioned such that the second zone **24** is within the field of view. The control unit **20** selectively actuates/deactuates the acoustic transducer **16** to cause separation of the blood cells and plasma into the first zone **22** and the second zone **24**. Then, in some embodiments, the control unit **20** actuates the reader **18** to capture information indicative of at least one parameter of the plasma. The information captured by the reader **18** is then transferred to the control unit **20** to determine a degree of hemolysis within the sample of blood. The control unit **20** can be constructed of circuitry and/or a combination of circuitry and software.

[0036] When the reader **18** is the optical reader, the degree of hemolysis can be determined by the control unit **20** based upon a colorimetric analysis of the sample. That is, when the sample is devoid of hemolysis and is illuminated with white light, the plasma will be substantially devoid of any color, i.e., the sample will be transparent. When hemolysis has occurred within the sample, the plasma will be pink when the plasma is illuminated with white light. By correlating the color of the plasma with predetermined colors indicative of an extent of hemolysis occurring within other samples, the extent of hemolysis within the sample can be determined. Depending upon a color of a backdrop, and/or color of illumination of the plasma, colors detected by the reader **18** indicative of an extent of hemolysis may differ.

[0037] Information indicative of an extent of hemolysis within the sample can be used to determine whether the blood has hemolysis.

[0038] FIG. **3** illustrates a blood testing device **40** constructed in accordance with one embodiment of the present disclosure. The blood testing device **40** is provided with a fluid container such as a syringe **42** or vacutainer having a fluid reservoir **44** for containing blood. The blood testing device **40** may also be provided with an acoustic transducer **46**, an optical zone **48**, and a bar code **50** which identifies the contents of the syringe **42** and can be correlated to specific patients. The acoustic transducer **46** may be provided with any suitable shape, such as planar, arcuate, or the like. In some embodiments, the acoustic transducer **46** may be provided with a shape to match a shape of the optical zone **48**, or other section of the blood testing device **40** to be stimulated by the acoustic transducer **46**. In such an embodiment, the blood testing device **40** allows the blood to be

acoustically treated using the acoustic transducer **46** after which the blood may be analyzed using an optical reader or the human eye through the optical zone **48**. Once a degree of hemolysis within the sample of blood has been determined, a decision can be made whether or not to continue with further testing of the sample of blood.

[0039] Referring now to FIGS. **4A-4C**, shown therein is an embodiment of a blood testing device **70** and reader **71**. The blood testing device **70** has a fluid housing **72**, a fluid reservoir **74**, a fluid treatment area **75**, an optical zone **76**, a treatment window **78**, and a flow port **82**. The reader **71** has a reading device **80**, a flow path **82**, and an analysis unit **84**. In this embodiment of the blood testing device **70**, a blood sample is contained in the fluid housing **72** and directed along the flow path **82** into the fluid treatment area **75** where the blood sample is directed to flow past the treatment window **78**. The treatment window **78** is constructed of a material that functions to pass acoustic forces generated by an acoustic transducer into the blood sample contained within the fluid housing **72**.

[0040] The reading device **80** is part of the analysis unit **84** and is provided with an acoustic transducer (not shown) and an optical reader (not shown) which operate as described above to acoustically treat the blood sample. The acoustic transducer may be provided with a planar shape so as to mate with the treatment window **78** of the blood testing device **70**. The optical reader of the reading device **80** has a field of view directed to the optical zone **76** where the acoustically treated blood sample may be read. The analysis unit **84** actuates the acoustic transducer to acoustically treat the blood sample and move the blood cells away from the optical zone **76** such that only the plasma is visible in the optical zone **76**. Then the optical reader captures an image of the plasma and any backdrop and sends the image to the analysis unit **84** for colorimetric analysis as discussed above. The analysis unit **84** may be provided with further blood analysis features (not shown) such as blood gas analysis which may further analyze the blood sample after it passes through the flow path **82**. The reader **71** may be portable and have a housing **88** that includes a slot **90** sized and dimensioned to receive the blood testing device **70** such that the optical zone **76** is in the field of view of the optical reader and the treatment window overlaps with the acoustic transducer. The housing **88** can be provided in a variety of shapes such as in a shape of a hot dog bun, for instance. The analysis unit **84** can be supported in the housing **88** or be separate therefrom. For example, the reading device **80** can be provided with a wireless transceiver to communicate with the analysis unit **84**. The analysis unit **84** may be constructed and function in a similar manner as the control unit **20** discussed above.

[0041] FIG. **5** illustrates another variation of the reader **71** in which the fluid housing **72** of FIG. **4** may be inserted into the housing **88** so that the blood sample may be acoustically treated and read by the reading device **80** as described above.

[0042] Referring now to FIG. **6**, a lateral flow blood testing device **100** constructed in accordance with the present disclosure is shown having a fluid housing **102**, a first fluid reservoir **104**, and a fluid treatment module **106**. The fluid treatment module **106** of the lateral flow blood testing device **100** is provided with a lower portion **108**, an upper portion **110**, a second fluid reservoir **111**, a lateral flow strip **112**, a fluid channel **114**, a first fluid port **116**, a second fluid port **118**, an optical zone **120**, and a bar code **122**.

[0043] The lower portion **108** and the upper portion **110** of the fluid treatment module **106** are sealably connected to form the second fluid reservoir **111**. When the fluid treatment module **106** is connected to the fluid housing **102**, a blood sample may be transferred from the first fluid reservoir **104** to the second fluid reservoir **111**. Once in the second fluid reservoir **111**, a portion of the blood sample may be directed through the second fluid port **118** into the fluid channel **114** where the blood sample passes through the lateral flow strip **112**. Through capillary action (which may also be referred to as capillary flow), the lateral flow strip **112** causes the separation of undamaged blood cells and plasma in the blood sample as described more fully in U.S. patent application Ser. No. 15/317,748, the entirety of which is incorporated herein by reference. The plasma that has

passed through the lateral flow strip **112** may then be analyzed in the optical zone **120** to determine a degree of hemolysis using an optical reader as described above or human eyes.

[0044] To facilitate directing the blood sample into the second fluid port **118**, the first fluid port **116** may be temporarily sealed using a removable cap (not shown), for instance, that temporarily prevents movement of the blood sample through the first fluid port **116**. When the blood sample has been analyzed using the lateral flow blood testing device **100**, the cap may be removed and the blood sample may be allowed to pass through the first fluid port **116** to be used for further testing, for instance, as desired.

[0045] As described above, the bar code **122** may be used to identify the blood sample, the patient the blood sample belongs too, the test to be performed, and the like.

[0046] FIG. 7 illustrates a lateral flow blood testing device **140** having a fluid housing **142**, a fluid treatment module **144**, a first fluid reservoir **146**, and a second fluid reservoir **148**. The lateral flow blood testing device **140** is similar to the lateral flow blood testing device **100** described above, therefore, in the interest of brevity only the differences will be described herein. In the embodiment shown in FIG. 7, the fluid treatment module **144** and the fluid housing **142** are integrated to form the lateral flow blood testing device **140**.

[0047] FIG. 8 illustrates a top plan view of a fluid treatment module **160** similar to fluid treatment modules **106** and **144**. In this embodiment, the fluid treatment module **160** is provided with a fluid channel **162** that connects a first fluid port **164** with a second fluid port **166** to direct a flow of a blood sample into a fluid channel **168** for separation of the blood sample into at least two constituent parts as described above. The fluid channel **168** houses the lateral flow strip **112** that functions to separate the undamaged blood cells from the plasma so that the plasma and any color caused by damaged blood cells is visible in the optical zone **120**.

[0048] Referring now to FIG. 9, shown therein is another embodiment of a lateral flow blood testing device **180** having a fluid housing **182**, a fluid reservoir **184**, and a fluid treatment module **186**. The fluid treatment module **186** of the lateral flow blood testing device **180** is provided with a lower portion **188**, an upper portion **190**, a second fluid reservoir **192**, and a lateral flow membrane **194**.

[0049] The lower portion **188** and the upper portion **190** of the fluid treatment module **186** are sealably connected to form the second fluid reservoir **192**. When the fluid treatment module **186** is connected to the fluid housing **182**, a blood sample may be transferred from the first fluid reservoir **184** to the second fluid reservoir **192**. As the blood sample is transferred from the first fluid reservoir **184** to the second fluid reservoir **192** the blood sample passes through the lateral flow membrane **194** and the blood sample is separated into at least two constituent parts, i.e., undamaged blood cells remain in the first fluid reservoir **184** and plasma with any damaged blood cells pass through the lateral flow membrane **194** and into the second fluid reservoir **192**.

[0050] At least the lower portion **188** of the fluid treatment module **186** is constructed of an optically clear material which allows the plasma that has passed through the lateral flow membrane **194** to be colorimetrically analyzed in the second fluid reservoir **192** using an optical reader as described above or human eyes.

[0051] Also shown in FIG. 9 is a probe **196** which may be attached to or part of a blood analysis machine (not shown) such as a blood gas analyzer. Where whole blood is needed for analysis, the probe **196** may be passed through a fluid port **198** in the fluid treatment module **186**, through the second fluid reservoir **192**, and through the lateral flow membrane **194** into the first fluid reservoir **184** where the blood sample has not been separated.

[0052] FIG. 10 illustrates another version of a lateral flow blood testing device **200** having a fluid housing **202**, a fluid treatment module **204**, a first fluid reservoir **206**, a second fluid reservoir **208**, a lateral flow membrane **210**, and a fluid port **214**. The lateral flow blood testing device **200** is similar to the lateral flow blood testing device **180** described above, therefore, in the interest of brevity only the differences will be described herein. In the embodiment shown in FIG. 10, the

fluid treatment module **204** and the fluid housing **202** are integrated into a unitary structure, rather than removably connected to form the lateral flow blood testing device **200**.

[0053] A probe **212** is also shown which may be attached to or part of a blood analysis machine (not shown) such as a blood gas analyzer. Where whole (unseparated) blood is needed for analysis, the probe **212** may be passed through the fluid port **214** in the fluid treatment module **204**, through the second fluid reservoir **206**, and through the lateral flow membrane **210** into the first fluid reservoir **206** where the blood sample has not been separated.

[0054] Referring now to FIGS. **11-13**, shown therein is yet another version of a blood testing device **240** similar to the blood testing device **70** shown in FIG. **4** that can be read by the reader **71**. In the interest of brevity, only the differences will be described in detail herein. The blood testing device **240** is provided with a fluid housing **242** a fluid reservoir **244**, a fluid treatment area **245**, an optical zone **246**, a treatment window **248**, a flow path **252**, a gate **254** having a port **256**, and a gate guide channel **258**.

[0055] When the gate **254** is in a first position (shown in FIG. **12**), the flow path **252** is restricted such that a blood sample in the flow path **252** stops in the optical zone **246** where the sample may be acoustically treated to move undamaged blood cells away from the plasma adjacent to the optical zone **246** and read with the reader **71** as described above.

[0056] Once the blood sample has been analyzed the gate **254** may be moved to a second position (shown in FIG. **13**), which moves the port **256** into the flow path **252** allowing the blood sample to pass.

[0057] Referring now to FIGS. **14-16**, shown therein is yet another version of a blood testing device **280** similar to the blood testing devices **70** and **240** shown in FIGS. **4** and **11-13**, respectively that can be read by the reader **71**. In the interest of brevity, only the differences will be described in detail herein. The blood testing device **280** is provided with a fluid housing **282** a fluid reservoir **284**, a fluid treatment area **285**, an optical zone **286**, a treatment window **288**, a first flow path **292**, a second flow path **294**, a gate **296** having a port **298**, and a gate guide channel **300**.

[0058] When the gate **296** is in a first position (shown in FIG. **15**), a blood sample is directed into the first flow path **292** such that the blood sample stops in the optical zone **286** where the blood sample may be acoustically treated and read with the reading device **80** as described above.

[0059] Once the blood sample has been analyzed the gate **296** may be moved to a second position (shown in FIG. **16**), which moves the port **298** into the second flow path **294** allowing the blood sample to pass through the second flow path **294**.

[0060] From the above description, it is clear that the inventive concepts disclosed herein is well adapted to carry out the objects and to attain the advantages mentioned herein as well as those inherent in the inventive concepts disclosed herein. While presently preferred embodiments of the inventive concepts disclosed herein have been described for purposes of this disclosure, it will be understood that numerous changes may be made which will readily suggest themselves to those skilled in the art and which are accomplished within the scope and coverage of the inventive concepts disclosed and claimed herein.

Claims

1. A blood testing assembly, comprising: a blood collection device, comprising: a first housing constructed of a fluid impermeable material and being partially constructed of a cylindrical shape having an internal cavity configured to receive blood, the first housing being a syringe or vacutainer; a treatment area supported by the first housing and defining a flow path in fluid communication with the internal cavity, the treatment area comprising: a treatment window adjacent to the flow path; an optical zone formed in the treatment window, the optical zone constructed of a material that allows a colorimetric analysis of a blood sample positioned within the flow path, and located adjacent to the optical zone; and a reader, comprising: a second housing

having an opening and receiving at least a portion of the treatment area having the treatment window and the optical zone.

2. A method, comprising: passing blood from a housing of a blood collection device through a flow path and into a lateral flow strip within a treatment area supported by the housing, a portion of the housing having a cylindrical shape, the housing being a syringe or vacutainer; separating blood cells from plasma within a blood sample by the lateral flow strip in the treatment area supported by the housing into a first zone containing plasma and blood cells, and a second zone containing plasma and being substantially devoid of blood cells; and colorimetrically analyzing the plasma within the second zone to determine a degree of hemolysis within the blood sample.

3. The method of claim 2, wherein the housing is a first housing, and wherein the treatment area has a second housing having a first end and a second end, the second housing having a connector at the first end, and wherein the method further comprises connecting the connector to the first housing prior to passing blood through the flow path.

4. The method of claim 2, wherein passing blood from the housing through the flow path, includes passing the blood sample outwardly towards a treatment window adjacent to the flow path.

5. The method of claim 2, wherein colorimetrically analyzing the plasma within the second zone is defined further as actuating an optical reader to capture information indicative of at least one parameter of the plasma.

6. The blood testing assembly of claim 1, wherein the treatment area includes a third housing having a first end and a second end, the third housing having a first connector at the first end and connected to the first housing, wherein the flow path extends through the first connector.

7. The blood testing assembly of claim 6, wherein the third housing has an outer peripheral wall supporting the treatment window, and wherein the flow path extends from the first connector outwardly to the treatment window.

8. The blood testing assembly of claim 6 wherein the third housing has a second end opposite to the first end, and wherein the third housing has a second connector on the second end.

9. The blood testing assembly of claim 1, further comprising a plunger positioned within the first housing such that the plunger and the first housing form the syringe.

10. The blood testing assembly of claim 1, wherein the reader further comprises an optical reader having a field of view overlapping the optical zone.

11. The blood testing assembly of claim 10, wherein the reader further comprises a control unit having circuitry configured to actuate the optical reader to capture information indicative of at least one parameter of plasma.

12. The blood testing assembly of claim 11 wherein the circuitry of the control unit is operable to colorimetrically analyze the information to determine a degree of hemolysis.

13. The blood testing assembly of claim 11, wherein the optical reader further comprises at least one of a camera and a photospectrometer.

14. The blood testing assembly of claim 1, wherein the reader further comprises: an acoustic transducer positioned inside the second housing and adjacent to the opening, the acoustic transducer configured to selectively generate acoustic forces directed through the treatment window and into the blood sample; and a control unit comprising circuitry configured to selectively actuate and deactivate the acoustic transducer.

15. A blood collection device, comprising: a housing constructed of a fluid impermeable material and being partially constructed of a cylindrical shape having an internal cavity configured to receive blood, the housing being a syringe or a vacutainer; and a treatment area supported by the housing and defining a flow path in fluid communication with the internal cavity, the treatment area comprising: a treatment window adjacent to the flow path; and an optical zone formed in the treatment window, the optical zone constructed of a material that allows a colorimetric analysis of a blood sample positioned within the flow path, and located adjacent to the optical zone.

16. The blood collection device of claim 15, wherein the housing is a first housing, and wherein the

treatment area includes a second housing having a first end and a second end, the second housing having a first connector at the first end and connected to the first housing, wherein the flow path extends through the first connector.

17. The blood collection device of claim 16, wherein the second housing has an outer peripheral wall supporting the treatment window, and wherein the flow path extends from the first connector outwardly to the treatment window.

18. The blood collection device of claim 16 wherein the second housing has a second end opposite to the first end, and wherein the second housing has a second connector on the second end.

19. The blood collection device of claim 16, further comprising a plunger positioned within the housing such that the plunger and the housing form the syringe.

20. The blood collection device of claim 15, further comprising a plunger positioned within the housing such that the plunger and the housing form the syringe.

21. A method, comprising: separating blood cells from plasma within a blood sample with acoustic treatment into a first zone containing plasma and blood cells, and a second zone containing plasma and being substantially devoid of blood cells; and analyzing the plasma within the second zone to determine a degree of hemolysis within the blood sample; and subsequent to analyzing the plasma, conducting a test of the blood sample by a blood analysis machine.

22. The method of claim 21, wherein conducting a test of the blood sample is defined further as conducting a blood gas analysis on the blood sample.

23. The method of claim 21, further comprising the step of inserting a probe attached to a blood analysis machine into a fluid reservoir containing the blood sample.

24. The method of claim 21, wherein conducting a test of the blood sample is defined further as conducting the test with blood analysis features of the blood analysis machine.

25. The method of claim 21, wherein analyzing the plasma within the second zone is defined further as colorimetrically analyzing the plasma within the second zone to determine the degree of hemolysis.

26. The method of claim 21, wherein analyzing the plasma within the second zone is defined further as analyzing the plasma within the second zone by analyzing an image of the plasma.

27. The method of claim 21, wherein analyzing the plasma within the second zone is defined further as analyzing the plasma within the second zone with a photospectrometer.
